B. Omkar and Last Class of Math

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: greedy, math, number theory | https://codeforces.com/problemset/problem/1372/B

In Omkar's last class of math, he learned about the least common multiple, or *LCM*. *LCM*(a, b) is the smallest positive integer x which is divisible by both a and b.

Omkar, having a laudably curious mind, immediately thought of a problem involving the *LCM* operation: given an integer n, find positive integers a and b such that $a + b = n$ and *LCM*(a, b) is the minimum value possible.

Can you help Omkar solve his ludicrously challenging math problem?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10$). Description of the test cases follows.

Each test case consists of a single integer n ($2 \le n \le 10^9$).

Output

For each test case, output two positive integers a and b, such that $a + b = n$ and *LCM*(a, b) is the minimum possible.

Example

Input
3 4 6 9

Output
2 2 3 3 3 6

Note

For the first test case, the numbers we can choose are 1, 3 or 2, 2. *LCM*(1, 3) = 3 and *LCM*(2, 2) = 2, so we output 2 2.

For the second test case, the numbers we can choose are 1, 5, 2, 4, or 3, 3. *LCM*(1, 5) = 5, *LCM*(2, 4) = 4, and *LCM*(3, 3) = 3, so we output 3 3.

For the third test case, *LCM*(3, 6) = 6. It can be shown that there are no other pairs of numbers which sum to 9 that have a lower *LCM*.